

MANILA, Philippines

WMO: 984250

Lat: 14.583N

Lon: 120.983E

Elev: 43

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.7	74.8	63.1	86.6	79.9	64.5	91.1	79.7	14.3	83.5	13.0	84.0	5.2	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.0	94.0	79.5	92.8	79.4	91.7	79.4	82.8	89.2	82.1	88.5	81.6	88.0	8.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.2	86.7	80.4	157.9	86.5	79.8	154.9	86.1	46.7	89.1	46.0	88.9	45.4	88.2	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.6	14.3	12.3	69.3	97.3	3.5	1.9	66.7	98.7	64.7	99.8	62.7	100.9	60.2	102.2		
			65.8	84.1	2.2	1.3	64.3	85.1	63.0	85.8	61.8	86.5	60.2	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.4	80.5	81.6	83.5	86.2	86.5	85.1	83.4	83.2	83.1	83.4	83.0	81.5
	DBStd	2.75	2.07	2.06	2.15	2.08	2.33	2.47	2.28	2.17	2.07	1.90	1.92	2.16
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12196	946	884	1038	1085	1132	1052	1035	1029	994	1035	990	976
	CDD65	6721	481	464	573	635	667	602	570	564	544	570	540	511
	CDH74	79912	4623	4817	6771	8501	9086	7786	6777	6666	6363	6853	6303	5366
CDH80	30930	1081	1398	2573	4225	4668	3561	2531	2401	2257	2553	2210	1472	
Wind	WSAvg	6.2	5.4	5.6	5.8	6.1	6.6	6.7	7.2	7.4	6.6	5.9	5.6	5.5
Precipitation	PrecAvg	84.0	0.4	0.2	0.5	0.9	5.0	12.3	17.2	18.2	14.1	8.0	5.2	2.2
	PrecMax	131.4	1.5	0.7	1.8	2.6	18.5	28.5	68.7	46.8	34.8	16.2	13.9	5.7
	PrecMin	60.3	0.0	0.0	0.0	0.0	0.6	3.8	3.9	8.3	1.4	1.2	0.5	0.0
	PrecStd	19.2	0.4	0.2	0.6	1.0	5.0	7.6	15.7	9.3	8.9	4.3	3.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	90.5	92.6	95.3	95.5	94.8	92.3	91.5	90.9	91.4	91.2	90.1
		MCWB	76.0	76.0	77.7	78.6	79.5	81.4	80.8	80.8	80.1	79.0	77.7	76.8
	2%	DB	87.3	88.9	91.1	93.7	94.2	92.7	90.3	89.8	89.6	89.8	89.5	88.3
		MCWB	75.7	76.0	77.3	78.4	79.9	80.8	80.5	80.7	80.2	79.0	78.2	76.9
	5%	DB	86.0	87.6	89.8	92.6	92.9	91.2	89.0	88.6	88.5	88.7	88.3	87.0
		MCWB	75.6	75.8	77.2	78.3	80.1	80.7	80.4	80.4	80.0	78.9	78.1	76.8
	10%	DB	84.7	86.3	88.5	91.3	91.5	89.8	87.9	87.4	87.4	87.6	87.2	85.7
		MCWB	75.2	75.6	76.8	78.4	80.3	80.5	80.1	80.1	79.8	78.8	78.0	76.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	80.0	81.9	82.6	83.4	83.7	83.4	82.7	82.4	82.3	81.7	80.1
		MCDB	85.5	85.7	87.2	90.3	90.7	91.1	88.8	87.5	87.5	88.3	87.4	86.7
	2%	WB	78.0	78.6	79.9	81.2	82.5	82.6	81.9	81.8	81.3	81.0	80.5	79.1
		MCDB	84.5	85.4	87.2	89.3	90.0	89.4	87.4	87.0	86.8	86.8	86.4	85.4
	5%	WB	77.2	77.6	79.0	80.4	81.8	81.8	81.2	81.1	80.8	80.5	79.7	78.3
		MCDB	83.7	84.6	86.8	88.8	89.4	88.4	86.8	86.5	86.6	86.2	85.8	84.4
	10%	WB	76.5	76.9	78.1	79.7	81.2	81.1	80.6	80.6	80.1	79.8	79.0	77.6
		MCDB	82.9	83.9	86.1	88.4	88.8	87.6	86.5	86.1	85.8	85.5	85.2	83.5
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.3	9.9	10.6	10.0	9.0	8.1	7.5	8.0	8.2	8.3	8.1
		MCDBR	9.5	10.8	11.4	11.9	11.7	10.6	9.9	9.3	9.5	9.6	9.8	9.6
		MCWBR	4.4	4.7	4.8	4.8	4.8	4.3	4.2	4.1	4.1	4.2	4.2	4.2
	5% WB	MCDBR	8.4	9.3	10.2	10.6	10.1	9.5	8.7	8.2	8.7	8.4	8.6	8.5
		MCWBR	5.0	5.1	5.3	5.2	4.9	4.7	4.5	4.3	4.3	4.4	4.7	4.5
Clear-Sky Solar Irradiance	taub	0.421	0.419	0.451	0.477	0.476	0.468	0.464	0.472	0.468	0.478	0.439	0.440	
	taud	2.418	2.408	2.321	2.253	2.286	2.323	2.335	2.304	2.311	2.262	2.388	2.370	
	Ebn at Noon	277	283	276	266	263	262	264	264	267	262	269	267	
	Edh at Noon	36	38	42	45	43	41	41	42	42	43	37	37	
All-Sky Solar Radiation	RadAvg	1342	1587	1792	2010	1817	1629	1437	1386	1473	1418	1330	1198	
	RadStd	81	135	121	107	162	164	184	143	148	116	118	103	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.58	+0.85	N/A	+0.81	+0.72	+0.97	N/A	N/A	+284	+284

Nomenclature: See separate page